DESIGN AND ANALYSIS OF ALGORITHMS

(Practical 1)

SOURCE CODE

```
import time

def fib_recursive(n):
    if n <= 1:
        return n
    return fib_recursive(n-1) + fib_recursive(n-2)

def fib_iterative(n):
    if n <= 1:
        return n
    a, b = 0, 1
    for _ in range(2, n+1):
        a, b = b, a+b
    return b

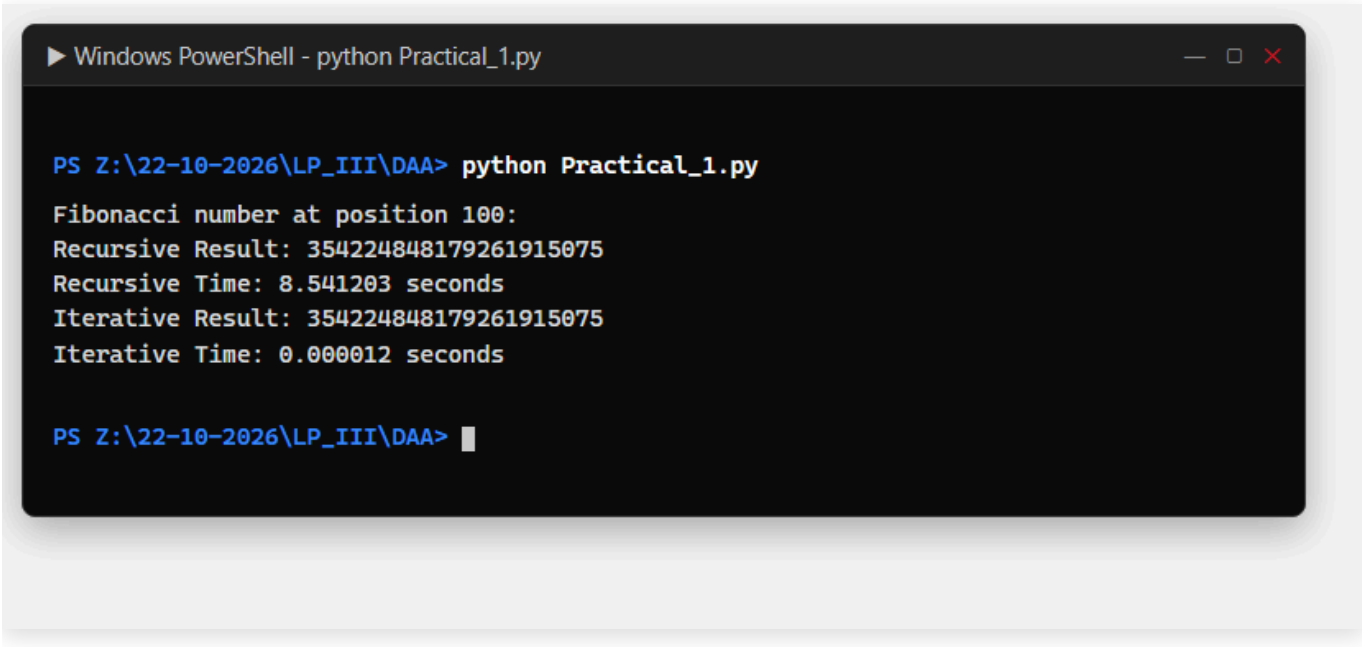
n = 100

print(f"Fibonacci number at position {n}:")

start = time.time()
result_recursive = fib_recursive(n)
end = time.time()
print(f"Recursive Result: {result_recursive}")
print(f"Recursive Time: {end - start:.6f} seconds")

start = time.time()
result_iterative = fib_iterative(n)
end = time.time()
print(f"Iterative Result: {result_iterative}")
print(f"Iterative Time: {end - start:.6f} seconds")
```

OUTPUT


Windows PowerShell - python Practical_1.py
PS Z:\22-10-2026\LP_III\DAA> python Practical_1.py
Fibonacci number at position 100:
Recursive Result: 354224848179261915075
Recursive Time: 8.541203 seconds
Iterative Result: 354224848179261915075
Iterative Time: 0.000012 seconds
PS Z:\22-10-2026\LP_III\DAA>